$$f(j,k) = \alpha'(b_j - b_{j+1} + b_{k-1} - b_k) + \alpha' c(\delta_{jp} + \delta_{kq} - \delta_{kp} - \delta_{jq})$$

$$r_j = \tilde{r}_j + c\alpha'(D_{jq} - D_{jp} + \delta_{jq} - \delta_{jp})$$

from?

$$f(j,k) = \alpha'(b_j - b_{j-1} + b_{k-1} - b_k) + \alpha' c(\delta_{jp} + \delta_{kq} - \delta_{kp} - \delta_{jp})$$

$$r_j := \alpha'(b_{j-1} - b_j + c\,\delta_{jq} - c\,\delta_{jp}) \quad \boxed{?}$$

$$r_j = \alpha'(b_{j-1} - b_j + c(\delta_{jq} - \delta_{jp}))$$

$$r_j = \alpha'(b_{j-1} - b_j) + c\alpha'(\delta_{jq} - \delta_{jp})$$

$$r_j = \underbrace{\tilde{r}_j + c\alpha'(D_{jq} - D_{jp}) + c\alpha'(\delta_{jq} - \delta_{jp})}_{} \checkmark$$

Test against $f(j,k)$ later

$$\tilde{\tilde{r}}_j := \tilde{r}_j + c(D_{jq} - D_{jp})$$

$$c = 1 - (b_p + b_{q-1} - b_{p-1} - b_q)\alpha'$$

$$\tilde{\tilde{r}}_q = \tilde{r}_q + c\alpha'(D_{qq} - D_{qp})$$

$$\tilde{\tilde{r}}_p = \tilde{r}_p + c\alpha'(D_{pq} - D_{pp})$$

Note! Diff def.

From?

$$c = 1 - \alpha'(b_p + b_{q-1} - b_{p-1} - b_q) = 1 - \alpha'(b_p - b_{p-1} - b_q + b_{q-1})$$

$$= 1 + \alpha'(b_{p-1} - b_p) - \alpha'(b_{q-1} - b_q)$$

$$= 1 + (\tilde{\tilde{r}}_p - \tilde{\tilde{r}}_q) = 1 - (\tilde{\tilde{r}}_q - \tilde{\tilde{r}}_p)$$

$$c = 1 + (\tilde{\tilde{r}}_p - \tilde{\tilde{r}}_q) = 1 + \left(\tilde{r}_p + c\alpha'(D_{pq} - D_{pp})\right) - \left(\tilde{r}_q + c\alpha'(D_{qq} - D_{qp})\right)$$

$$c = 1 + (\tilde{r}_p - \tilde{r}_q) + c\alpha'(D_{pq} - D_{pp}) - c(D_{qq} - D_{qp})$$

$$c = 1 + (\tilde{r}_p - \tilde{r}_q) + c\alpha'\left[D_{pq} - D_{pp} - D_{qq} + D_{qp}\right]$$

$$c\left[1 - \alpha'\left[D_{pq} - D_{pp} - D_{qq} + D_{qp}\right]\right] = 1 + (\tilde{r}_p - \tilde{r}_q)$$

$$c = \frac{1 + (\tilde{r}_p - \tilde{r}_q)}{\left[1 - \alpha'\left[D_{pq} - D_{pp} - D_{qq} + D_{qp}\right]\right]}$$

Equations tested:

test_Ka_y: $Ka = y$

test_Ka_y: $Ka = y$ exception case

test_B_ij: $\alpha' B_{ij} = \dfrac{\sinh((n-i+j)\lambda)\sinh((i+j)\lambda)}{\sinh(\lambda)\sinh(n\lambda)} = (I - \alpha' A)^{-1}$

test_b_mult: $b = 1 - c\alpha'(e_p + e_{q-1} - e_{p-1} - e_q) = b_array$

from inverse

test_b_i: $b_i = \tilde{b} - c\alpha'(B_{ip} + B_{iq-1} - B_{ip-1} - B_{iq})$

test_c_analytical $c = 1 - \alpha'(b_p + b_{q-1} - b_{p-1} - b_q) = c[1]$

test_f_j_k:

$$K_{jk} a = \alpha'(b_j - b_{j-1} + b_{k-1} - b_k) + \alpha' c(\delta_{jp} + \delta_{kq} - \delta_{kp} - \delta_{jq})$$

test_rank_mult_exp: $r_j = \alpha'(b_{j-1} - b_j + c(\delta_{jq} - \delta_{jp}))$

$$K_{jk} a = r_k - r_j$$

test B_{ij}: all cases of analytical

$$\| $$

$$\alpha'(B_{i-1j} + B_{ij-1} - B_{ij} - B_{i-1j-1})$$

test_alpha_b:

$$\alpha'(b_{j+1} - b_j) = \tilde{r}_j + c\,\alpha'(D_{jq} - D_{jp})$$

test rank_and_rank_til:

$i \wedge j$

$$r_j - \alpha'(b_{j-1} - b_j + c(\delta_{jq} - \delta_{jp})) = \underbrace{\tilde{r}_j + c\alpha'(D_{jq} - D_{jp} + \delta_{jq} - \delta_{jp})}$$

and

$$R_{ik} = r_k - r_j$$

$$\left[\tilde{r}_j + \tilde{r}_{j\,diff}\right]$$

test_rank_d_til: $c = 1 + \tilde{r}_p + r_q$

test_c_double_til: $c = \dfrac{1 + \tilde{r}_p - \tilde{r}_q}{1 - \alpha'(D_{pq} - D_{pp} - D_{qq} + D_{qp})}$ Type this up.

→ plot as func of α, and overall c

check case 3

check D_{ij}

$$\alpha = 1 - 2\frac{\delta_0}{\delta_S}$$

$0 < \alpha' < 0.5$

$\dfrac{0.1}{\text{denominator}}$

- Analysis → Theor. and
- behavioral predictions

Section 5.4c

$$D_{ij} = \begin{cases} i=j: & -1 + \dfrac{\cosh\left(\left(i\lambda_j - \frac{1}{2}\right)\lambda\right)\cosh\left(\left(n-\left(i\nu_j - \frac{1}{2}\right)\right)\lambda\right)\tanh(\lambda/2)}{\sinh(\lambda n)} \\ i \neq j: & \dfrac{\cosh\left(\left(i\lambda_j - \frac{1}{2}\right)\lambda\right)\cosh\left(\left(n-\left(i\nu_j - \frac{1}{2}\right)\right)\lambda\right)\tanh(\lambda/2)}{\sinh(\lambda n)} \end{cases}$$

Powerpoint relations:

ok_ok, Δr_orig - r_{jk}, yet more prominent

$r_{orig} - r_j =$

Graph: $r_{orig} - (-r_j) = -\tilde{r}_j - (-\tilde{r}_j - c\alpha'(D_{jq} - D_{jp} + \delta_{jq} - \delta_{jp}))$

$r_{orig} = -\tilde{r}_j$

$$= \underbrace{c\alpha'(D_{jq} - D_{jp} + \delta_{jq} - \delta_{jp})}_{c\alpha'(\delta_{jp} - \delta_{jp})}$$

$\alpha'(\ldots)$

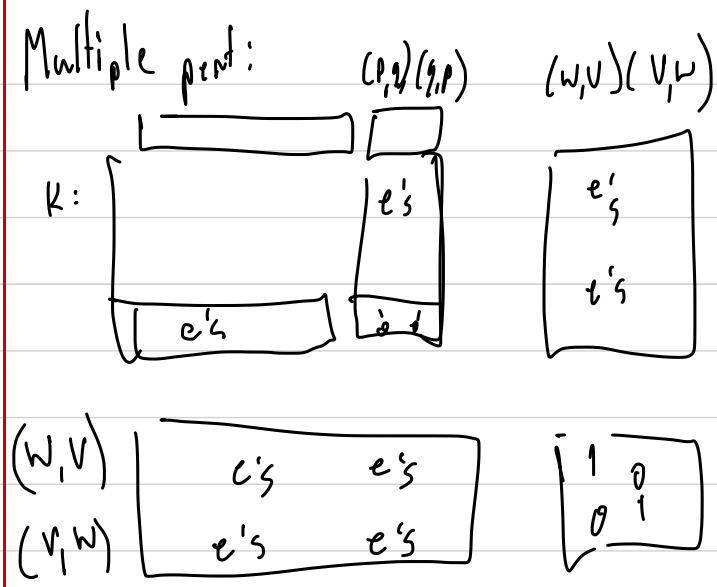
↳ should be a general diff

$q=2,\ p=5$

	D_{jq}	D_{jp}
0.99	0.109...	0.15
0.3	0.028	0.07

Interesting?

Multiple pert:

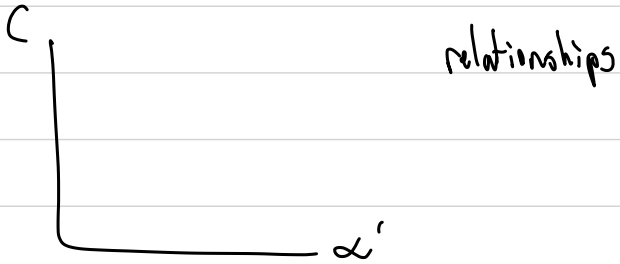


$K_0 \to 0.5$ is interesting?

To what degree is the brain actually doing regression while learning?
One-shot learning? → memorization
How does it relate to our model of memorization?

Kemp & Tenenbaum 2008 graph

Analytical solutions



relationships

Depending on what

For D_{ij}, if $i \leq q$, trend changes for α'vs. D_{ij}.

Conj values:

Become better.

$\boxed{\text{Plot}}$ [Ranked exception & K_{-0} *]

Memorization term

[ ]

$$D_{ij} = \frac{\left(\sinh((i-1)\lambda) - \sin(i\lambda)\right)\left(\sinh((n-j)\lambda) - \sinh((n+1-j)\lambda)\right)}{\sinh(\lambda)\sinh(n\lambda)}$$

$x = i-1 \quad y = i \quad x+y = 2i-1 \quad x-y = -1$; $x = n-j \quad y = n+1-j = n-(j-1)$

$x+y = -(2j-1) \quad \frac{x+y}{2} = -j+\frac{1}{2} = -(j-\frac{1}{2})$

$$4\left[\cosh\left(i-\frac{1}{2}\right)\sinh\left(-\frac{1}{2}\lambda\right)\right]\left[\cosh\left(\left(n-\left(j-\frac{1}{2}\right)\right)\lambda\right)\sinh\left(-\frac{\lambda}{2}\right)\right]$$

$$= \frac{4\overbrace{\sinh^2\left(-\frac{\lambda}{2}\right)}^{*}\left[\cosh\left(i-\frac{1}{2}\right)\cosh\left(n-\left(j-\frac{1}{2}\right)\lambda\right)\right]}{\sinh(\lambda)\sinh(n\lambda)}$$

$$\frac{\sinh\left(-\frac{\lambda}{2}\right)}{\sinh(\lambda)} = \frac{\quad}{\sinh(\lambda)}$$

$\sinh\left(-\frac{\lambda}{2}\right) = 2\sinh\left(-\frac{\lambda}{4}\right)\cosh\left(-\frac{\lambda}{4}\right)$ using $\sinh(2x) = 2\sinh(x)\cosh(x)$

$$B_{ij} = \frac{\sinh((n-|i-j|)\lambda)\sinh((i+j)\lambda)}{\sinh(\lambda)\sinh(n\lambda)}$$

Case 3 $i=j$

$$D_{ij} = \alpha'\left(2B_{(i-1)j} - B_{ij} - B_{(i-1)(j-1)}\right)$$

$$= \frac{2\sinh((n-j)\lambda)\sinh(i\lambda) - \sinh((n-i)\lambda)\sinh(i\lambda) - \sinh((n-i-1)\lambda)\sinh((i-1)\lambda)}{\sinh(\lambda)\sinh(n\lambda)}$$

$=$ use cosh trick

$=$

1st part

Moves done:

- $\sinh 2x = 2\sinh x \cosh x$

- $\cosh x - \cosh y = 2\sinh\left(\frac{x+y}{2}\right)\sinh\left(\frac{x-y}{2}\right)$

- $\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$

- $\bar{a}_j - \bar{a}_{j-1}$ ∴ similar terms cancel

2nd part

- $\cosh\left(\frac{x}{2}\right) = \sqrt{\frac{\cosh x + 1}{2}}$
- $\sinh\left(\frac{x}{2}\right) = \frac{\sinh x}{\sqrt{2}\sqrt{\cosh x + 1}}$

} use to cancel $\frac{\sinh x}{\cosh\left(\frac{x}{2}\right)}$

- $\sinh(\lambda) = \overbrace{2\sinh\left(\frac{\lambda}{2}\right)\cosh\left(\frac{\lambda}{2}\right)}^{\text{cancel later.}}$

- $\overbrace{\cosh\left(\left(\frac{n}{2}+1-j\right)\lambda\right) - \cosh\left(\left(\frac{n}{2}-j\right)\lambda\right)}^{\text{similar terms}}$
$= \sinh\left(\left(\left(\frac{n+1}{2}\right)-j\right)\lambda\right)\sinh\left(\frac{\lambda}{2}\right)$

eliminated later

Before w/ recombining

- $2AB - CB - AD = A(B-D) - B(A-C)$

- $(B-C) \Rightarrow \sinh(a)\sinh(b)$
  $(A-C) \Rightarrow \sinh(a)\sinh(c)$

$2AB - CD - AD = A(B-D) - B(A-C) = \sinh(a)(WX - YZ)$

Num-1: Idea 1: Find $\cosh\left(\left(\frac{n+1}{2}\right)\lambda\right)$ Didn't work

$$\sinh\left(\overbrace{\left(\frac{n+1}{2}-q\right)}^{a}\lambda\right)-\sinh\left(\overbrace{\left(\frac{n+1}{2}-p\right)}^{b}\lambda\right)$$

$$= 2\cosh\left(\left(\frac{n+1}{2}-\frac{p+q}{2}\right)\lambda\right)\sinh\left(\frac{p+q}{2}\lambda\right)$$

$$\cosh\left(\left(\frac{n+1}{2}\right)\lambda+\left(\frac{p+q}{2}\right)\lambda\right)$$

$$= \cosh\left(\left(\frac{n+1}{2}\right)\lambda\right)\cosh\left(\frac{p+q}{2}\lambda\right) + \sinh\left(\frac{n+1}{2}\lambda\right)\sinh\left(\frac{p+q}{2}\lambda\right)$$

$$= 2\sinh\left(\frac{p+q}{2}\lambda\right)\left[\cosh\left(\left(\frac{n+1}{2}\right)\lambda\right)\cosh\left(\frac{p+q}{2}\lambda\right)+\sinh\left(\frac{n+1}{2}\lambda\right)\sinh\left(\frac{p+q}{2}\lambda\right)\right]$$

Num_d: $\sinh\left(\frac{n+1}{2}\lambda\right)-\sinh\left(\frac{n-1}{2}\lambda\right)=2\cosh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right)$

Num: $$\frac{2\cosh\left(\left(\frac{n+1}{2}-\frac{p+q}{2}\right)\lambda\right)\sinh\left(\frac{p+q}{2}\lambda\right)+2\cosh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right)}{\sinh\left(\frac{n+1}{2}\lambda\right)-\sinh\left(\frac{n-1}{2}\lambda\right)}$$

Symp $$\overset{?}{=}\frac{2\cosh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right)+\sinh\left(\left(\frac{n}{2}-q+\frac{1}{2}\right)\lambda\right)-\sinh\left(\left(\frac{n}{2}-p+\frac{1}{2}\right)\lambda\right)}{\sinh\left(\frac{n+1}{2}\lambda\right)-\sinh\left(\frac{n-1}{2}\lambda\right)}$$

Idea 2: Combine w/ 1

Num:

a b

$$\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right) + \sinh\left(\left(\frac{n+1}{2} - q\right)\lambda\right) - \sinh\left(\left(\frac{n+1}{2} - p\right)\lambda\right)$$

Idea:
Change $\sinh\left(\left(\frac{n-1}{2}\right)\lambda\right)$ on bottom?

$$\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\left(\frac{n+1}{2} - p\right)\lambda\right)$$

$$= 2\cosh\left(\left(\frac{n+1}{2} - \frac{p}{2}\right)\lambda\right)\sinh\left(\frac{p}{2}\lambda\right)$$

$$-\left[\sinh\left(\frac{n-1}{2}\lambda\right) - \sinh\left(\left(\frac{n+1}{2} - q\right)\lambda\right)\right]$$

$$= -2\cosh\left(\left(\frac{n}{2} - \frac{q}{2}\right)\lambda\right)\sinh\left(\left(\frac{q}{2} - \frac{1}{2}\right)\lambda\right)$$

q & p are switcheable depending on what you want

Do something w/ $\frac{q}{2}$?

$\cosh(x-y) = \cosh x \cosh y - \sinh x \sinh y$

$\sinh(x-y) = \sinh x \cosh y - \cosh x \sinh y$

$\Rightarrow$ $a - b$

$$\Rightarrow 2\cosh\left(\left(\frac{n+1}{2} - \frac{p}{2}\right)\lambda\right)\sinh\left(\frac{p}{2}\lambda\right) - 2\cosh\left(\left(\frac{n}{2} - \frac{q}{2}\right)\lambda\right)\sinh\left(\left(\frac{q}{2} - \frac{1}{2}\right)\lambda\right)$$

$$\cosh\left(\left(\frac{n+1}{2}\right)\lambda - \left(\frac{p}{2}\right)\lambda\right) = \cosh\left(\left(\frac{n+1}{2}\right)\lambda\right)\cosh\left(\frac{p}{2}\lambda\right) - \sinh\left(\left(\frac{n+1}{2}\right)\lambda\right)\sinh\left(\frac{p}{2}\lambda\right)$$

$$\cosh\left(\left(\frac{n}{2}\right)\lambda - \left(\frac{p}{2}\right)\lambda\right) = \cosh\left(\left(\frac{n}{2}\right)\lambda\right)\cosh\left(\frac{p}{2}\lambda\right) - \sinh\left(\left(\frac{n}{2}\right)\lambda\right)\sinh\left(\frac{p}{2}\lambda\right)$$

$$\sinh\left(\left(\frac{q}{2}\right)\lambda - \left(\frac{1}{2}\right)\lambda\right) = \sinh\left(\frac{q}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right) - \cosh\left(\frac{q}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right)$$

$$\Rightarrow 2\sinh\left(\frac{p}{2}\lambda\right)\left[\cosh\left(\left(\frac{n+1}{2}\right)\lambda\right)\cosh\left(\frac{p}{2}\lambda\right) - \sinh\left(\left(\frac{n+1}{2}\right)\lambda\right)\sinh\left(\frac{p}{2}\lambda\right)\right]$$

$$- 2\left[\sinh\left(\frac{q}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right) - \cosh\left(\frac{q}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right)\right] \times$$

$$\left[\cosh\left(\left(\frac{n}{2}\right)\lambda\right)\cosh\left(\frac{p}{2}\lambda\right) - \sinh\left(\left(\frac{n}{2}\right)\lambda\right)\sinh\left(\frac{p}{2}\lambda\right)\right]$$

$$\cosh\left(\frac{n}{2}\lambda + \frac{\lambda}{2}\right) = \cosh\left(\frac{n}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right) + \sinh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right)$$

$$\Rightarrow 2\sinh\left(\frac{p}{2}\lambda\right)\left[\left(\cosh\left(\frac{n}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right) + \sinh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right)\right)\cosh\left(\frac{p}{2}\lambda\right)\right.$$

$$\left. - \left(\sinh\left(\frac{n}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right) + \cosh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right)\right)\sinh\left(\frac{p}{2}\lambda\right)\right]$$

Weird middle term?

$$- 2\left[\sinh\left(\frac{q}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right) - \cosh\left(\frac{q}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right)\right] \times$$

$$\left[\cosh\left(\left(\frac{n}{2}\right)\lambda\right)\cosh\left(\frac{p}{2}\lambda\right) - \sinh\left(\left(\frac{n}{2}\right)\lambda\right)\sinh\left(\frac{p}{2}\lambda\right)\right]$$

Try subtraction formulas

$$\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right) + \sinh\left(\left(\frac{n+1}{2} - q\right)\lambda\right) - \sinh\left(\left(\frac{n+1}{2} - p\right)\lambda\right)$$

$= a - b$

$$= 2\cosh\left(\left(\frac{n+1}{2} - \frac{p}{2}\right)\lambda\right)\sinh(p\lambda) - 2\cosh\left(\left(\frac{n}{2} - \frac{q}{2}\right)\lambda\right)\sinh(q\lambda)$$

Idea: Try eliminating $\sinh\left(\frac{n-1}{2}x\right)$

$$\sinh\left(\frac{n+1}{2}x\right) = \sinh\left(\frac{n}{2}x + \frac{1}{2}x\right) = \sinh\left(\frac{n}{2}x\right)\cosh\left(\frac{x}{2}\right) + \cosh\left(\frac{n}{2}x\right)\sinh\left(\frac{x}{2}\right)$$

$$\sinh\left(\frac{n-1}{2}x\right) = \sinh\left(\frac{n}{2}x - \frac{1}{2}x\right) = \sinh\left(\frac{n}{2}x\right)\cosh\left(\frac{x}{2}\right) - \cosh\left(\frac{n}{2}x\right)\sinh\left(\frac{x}{2}\right)$$

$$\sinh\left(\frac{n+1}{2}x\right) - \sinh\left(\frac{n-1}{2}x\right) = 2\cosh\left(\frac{n}{2}x\right)\sinh\left(\frac{x}{2}\right)$$

Denominator

$$\alpha = 4\alpha' \frac{\tanh(\lambda/2)}{\sinh(\lambda n)} \left(\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right) \sinh\left(\left(n-\left(\frac{p+q-1}{2}\right)\right)\lambda\right) + \cosh\left(\left(n-\left(p-\tfrac{1}{2}\right)\right)\lambda\right) \sinh\left(\frac{p+q-1}{2}\lambda\right) \right)$$

$$\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right) \sinh\left(\left(n-\left(\frac{p+q-1}{2}\right)\right)\lambda\right) + \cosh\left(\left(n-\left(p-\tfrac{1}{2}\right)\right)\lambda\right) \sinh\left(\frac{p+q-1}{2}\lambda\right)$$

$$\sinh\left(\left(n-\left(\frac{p+q-1}{2}\right)\right)\lambda\right) = \sinh(n\lambda)\cosh\left(\frac{p+q-1}{2}\lambda\right) - \sinh\left(\left(\frac{p+q-1}{2}\right)\lambda\right)\cosh(n\lambda)$$

$$\sinh(\lambda n) = 2\sinh\left(\frac{\lambda}{2}n\right)\cosh\left(\frac{\lambda}{2}n\right)$$

$$\tanh\left(\frac{\lambda}{2}\right) = \frac{\sinh(\lambda/2)}{\cosh(\lambda/2)} =$$

$$\cosh\left(\left(n-\left(p-\tfrac{1}{2}\right)\right)\lambda\right) = \cosh(n\lambda)\cosh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) - \sinh(n\lambda)\sinh\left(\left(p-\tfrac{1}{2}\right)\lambda\right)$$

$$\Rightarrow \cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\left[\sinh(n\lambda)\cosh\left(\frac{p+q-1}{2}\lambda\right) - \sinh\left(\left(\frac{p+q-1}{2}\right)\lambda\right)\cosh(n\lambda)\right]$$

$$+ \sinh\left(\frac{p+q-1}{2}\lambda\right)\left[\cosh(n\lambda)\cosh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) - \sinh(n\lambda)\sinh\left(\left(p-\tfrac{1}{2}\right)\lambda\right)\right]$$

Almost $\sinh(n\lambda)\left[\cosh\left(\left(\left(\frac{p+q-1}{2}\right) - \left(p/q - \frac{1}{2}\right)\right)\lambda\right)\right]$

$\cosh(n\lambda)\left[\sinh\left(\left(\left(\frac{p+q-1}{2}\right) - \left(p/q - \frac{1}{2}\right)\right)\lambda\right)\right]$

skip this

$$= \sinh(n\lambda)\left[\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\frac{p+q-1}{2}\lambda\right) - \sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\left(p-\tfrac{1}{2}\right)\lambda\right)\right]$$

$$+ \cosh(n\lambda)\sinh\left(\frac{p+q-1}{2}\lambda\right)\left(\cosh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) - \cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\right)$$

$$= \sinh(n\lambda)\left[\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\frac{p+q-1}{2}\lambda\right) - \sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\left(p-\tfrac{1}{2}\right)\lambda\right)\right]$$

$$+ \cosh(n\lambda)\sinh\left(\frac{p+q-1}{2}\lambda\right)\left(2\sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\frac{p-q}{2}\lambda\right)\right)$$

$$= \sinh(n\lambda)\left[\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\frac{p+q-1}{2}\lambda\right) - \sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\left(p-\tfrac{1}{2}\right)\lambda\right)\right]$$

$$+3\cosh(n\lambda)\sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\frac{p-q}{2}\lambda\right)$$

New identities: $\cosh(a)\cosh(b) = \frac{1}{2}\left[\cosh(a)\cosh(b) + \sin(a)\sin(b) + \cosh(a)\cosh(b) - \sin(a)\sin(b)\right]$

$$= \frac{1}{2}\left(\cosh(a+b)\right)$$

Using the branch:

$$= \sinh(n\lambda)\left[\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\frac{p+q-1}{2}\lambda\right) - \sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\left(p-\tfrac{1}{2}\right)\lambda\right)\right]$$

$$+\cosh(n\lambda)\left[\sinh\left(\frac{p+q-1}{2}\lambda\right)\cosh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) - \cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\sinh\left(\left(\frac{p+q-1}{2}\right)\lambda\right)\right]$$

Identities.

$$\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\frac{p+q-1}{2}\lambda\right) = \frac{1}{2}\left[\cosh\left(\left(q-\tfrac{1}{2}+\frac{p+q-1}{2}\right)\lambda\right) + \cosh\left(\left(q-\tfrac{1}{2}-\frac{p+q-1}{2}\right)\lambda\right)\right]$$

$$= \frac{1}{2}\left[\cosh\left(\left(\frac{3q+p-2}{2}\right)\lambda\right) + \cosh\left(\left(\frac{q-p}{2}\right)\lambda\right)\right]$$

$$\sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) = \frac{1}{2}\left[\cosh\left(\left(p-\tfrac{1}{2}+\frac{p+q-1}{2}\right)\lambda\right) - \cosh\left(\left(\frac{p+q-1}{2}-\left(\frac{2p}{2}-\tfrac{1}{2}\right)\right)\lambda\right)\right]$$

$$= \frac{1}{2}\left[\cosh\left(\left(\frac{3p+q-2}{2}\right)\lambda\right) + \cosh\left(\left(\frac{q-p}{2}\right)\lambda\right)\right]$$

$$\sinh\left(\frac{p+q-1}{2}\lambda\right)\cosh\left(\left(p-\frac{1}{2}\right)\lambda\right) = \frac{1}{2}\left[\sinh\left(\frac{2p+q-\frac{3}{2}}{2}\lambda\right) + \sinh\left(\frac{q-\frac{1}{2}}{2}\lambda\right)\right]$$

$$\sinh\left(\left(\frac{p+q-1}{2}\right)\lambda\right)\cosh\left(\left(q-\frac{1}{2}\right)\lambda\right) = \frac{1}{2}\left[\sinh\left(\frac{2q+p-\frac{3}{2}}{2}\lambda\right) + \sinh\left(\frac{p-\frac{1}{2}}{2}\lambda\right)\right]$$

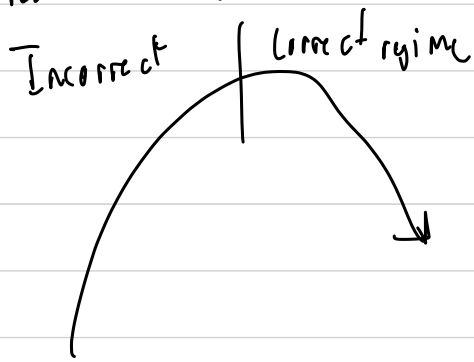
$$\alpha - 4\alpha' \frac{\tanh(\lambda/2)}{\sinh(\lambda n)} \left(\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right) \sinh\left(\left(n-\left(\frac{p+q-1}{2}\right)\right)\lambda\right) + \cosh\left(\left(n-\left(p-\tfrac{1}{2}\right)\right)\lambda\right) \sinh\left(\frac{p+q-1}{2}\lambda\right) \right)$$

$$1 = -2\alpha' \cosh(\lambda)$$

$$-2\alpha' \cosh(\lambda) - 4\alpha' \left(\text{etc...} \right)$$

$$= \alpha' \left(-2\cosh(\lambda) - 4(\text{etc...}) \right)$$

Mechanistic examination:



D_{ij}-diff vs. α'

D_{ij}

Checking $\bar{a}_i$ simplifier:

$$\bar{a}_i = \frac{b_i^{(1)} - b_i^{(2)}}{2\delta_s \alpha \sinh(\lambda) \sinh(\lambda n)}$$

$$r_j = \delta_6 \frac{1-\alpha}{2} \left(\bar{a}_j - \bar{a}_{j-1} \right)$$

$$b_i^{(1)} = \frac{num}{\sinh(\lambda 12)}$$

$$b_i^{(2)} = \frac{num}{\sinh(\lambda 12)}$$